

UNIVERSITY OF DHAKA

Department of Computer Science and Engineering



AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

Thesis Progress Report

Project Period: February 2026 - February 2027

Status Date: 6 September 2026

Submitted by

Md. Rakib Hossain

Roll: 88 Registration No.: 2020315665

Shah Jamal Islam

Roll: 81 Registration No.: 2020915641

Supervised by

Dr. Saifuddin Md. Tareeq

Professor

Department of Computer Science and Engineering

University of Dhaka

CSE 4114 - Project Work

4th Year 1st Semester Examination 2025 September 2026

Declaration Form

This report records the progress of the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. We declare that the implementation, experiments and analysis reported as completed were carried out by the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets and prior findings originating from other works are cited.

The document deliberately separates results obtained from synthetic data, public real data and future primary SME data. No synthetic result is presented as evidence of field effectiveness. This progress report is current to 6 September 2026; the thesis remains in progress until February 2027.

Md. Rakib Hossain
Candidate

Shah Jamal Islam
Candidate

Dr. Saifuddin Md. Tareeq
Supervisor

Abstract

AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

Small retail businesses in Bangladesh routinely create sales, inventory, payment and customer records, yet many lack an integrated mechanism for turning those records into forecasts and practical decisions. This thesis proposes B-SMART, a Bangladesh-focused framework that connects daily business operations with leakage-aware machine learning, explainability and constraint-aware action ranking. The intended outputs are SKU demand forecasts, future customer inactivity scores, customer segments, performance indicators and recommendations such as reorder or retention priorities.

As of 6 September 2026, a local end-to-end minimum viable product is implemented. It supports organization and branch isolation, catalogue and inventory management, sales, purchases, returns, expenses, ledgers, durable CSV/XLSX imports, model training, a FastAPI service and a React dashboard. The research pipeline includes supervised and unsupervised segmentation, LSTM and ARIMA forecasting with a seasonal-naive baseline, churn and return prediction, SHAP analysis, and chronological real-data validation. Six automated backend tests pass and the production frontend builds successfully.

Current experiments are real-data first. Under an identical daily-demand target, expanding-window selection and fixed chronological holdouts, dataset-specific models obtain 17.49% WAPE on a 1,826-day real Bangladeshi retailer series and 30.37% on UCI Online Retail II. On UCI customer snapshots, 90-day future-repeat prediction reaches ROC-AUC 0.749, PR-AUC 0.795 and 1.77x top-decile lift. A separate WFP/DAM Bangladesh price benchmark gives holdout WAPE between 3.14% and 7.85% for the winning method across four staple series; simple last-value forecasts win three. Controlled generated-data experiments are reported only as comparisons: development-only tuning selects XGBoost at 69.90% segment holdout accuracy (95% CI [67.0%, 72.6%]), synthetic churn reaches 2.67x lift, and return prediction remains weak (PR-AUC 0.083). These outcomes expose why leakage control, untouched tests and strong baselines matter more than selecting a model after observing favourable test accuracy.

These are prototype and external-validation results, not final field claims. The primary unfinished dependency is consenting invoice-level data from Bangladeshi SMEs. Work through February 2027 therefore focuses on ethics and data agreements, primary data collection, calibrated temporal evaluation, ablation and scalability studies, owner usability testing, business-impact simulation or pilot evaluation, and final thesis consolidation.

Keywords: Bangladesh SMEs, business analytics, demand forecasting, customer segmentation, churn prediction, explainable AI, decision support.

Acknowledgements

We express our sincere gratitude to our supervisor, Dr. Saifuddin Md. Tareeq, Professor, Department of Computer Science and Engineering, University of Dhaka, for defining the academic direction of this work and for emphasizing clear problem formulation, literature coverage, sequential methodology, scalability evaluation and honest reporting. We also thank the faculty and staff of the department for the facilities and guidance that support this project. Finally, we acknowledge the maintainers of the public datasets and open-source software used for reproducible method validation. Responsibility for the implementation, analysis and any remaining errors belongs to the authors.

Contents

Declaration Form	i
Abstract	ii
Acknowledgements	iii
List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 Motivations	1
1.2 Problem Statement	2
1.3 Objectives	2
1.3.1 General Objective	2
1.3.2 Specific Objectives	2
1.4 Contributions	3
1.5 Challenges	4
1.6 Scope and Current Report Boundary	4
1.7 Organization	4
2 Related Works	5
2.1 Background Study	5
2.1.1 Business Intelligence and Decision Support	5
2.1.2 Bangladeshi SME and Payment Context	5
2.1.3 Customer Analytics: RFM, CLV and Segmentation	6
2.1.4 Demand Forecasting	6
2.1.5 Churn, Future Repeat and Returns	7
2.1.6 Explainability, Leakage and Model Governance	7
2.2 Literature Review	7
2.2.1 Bangladesh and Business Analytics Foundations	7
2.2.2 Customer Analytics and Segmentation	8
2.2.3 Predictive Models and Forecasting	9
2.2.4 Churn, Evaluation Integrity and Explainability	10
2.3 Literature Matrix	11
2.4 Problem of Existing Systems	13
2.5 Research Gap and Positioning	13

3	Proposed Methodologies	15
3.1	Research Design	15
3.2	Research Questions and Hypotheses	15
3.3	Target Population and Data Acquisition	16
3.4	Ethics, Privacy and Data Governance	17
3.5	Canonical Data Model	17
3.6	Formal Problem Formulation	18
3.6.1	Demand Prediction	18
3.6.2	Future Customer Inactivity	18
3.6.3	Strategy Optimization	19
3.7	Preprocessing and Historical Feature Construction	19
3.8	Analytical Models	19
3.8.1	Segmentation and Classification	19
3.8.2	Forecasting	19
3.8.3	Churn and Return Risk	20
3.8.4	Explainability	20
3.9	The B-SMART Sequential Algorithm	20
3.10	Architecture and Sequential Flow	21
3.11	Worked Recommendation Example	23
3.12	Evaluation Protocol	23
3.12.1	Forecasting Metrics and Splits	23
3.12.2	Classification and Ranking Metrics	24
3.12.3	Clustering Evaluation	24
3.12.4	Scalability, Usability and Business Impact	24
4	Implementation	25
4.1	Implementation Status	25
4.2	Technology Stack	26
4.3	Repository and Layered Architecture	26
4.4	Operational Layer	27
4.4.1	Tenant and Branch Isolation	27
4.4.2	Catalogue, Inventory and Sales	27
4.4.3	Purchasing, Returns and Accounts	27
4.5	Durable Data Intake	27
4.6	Analytical Pipeline	28
4.7	Strategy Optimization Layer	29
4.8	API and User Interface	29
4.9	Verification and Build Status	29
4.10	Reproducibility	30
5	Experimental Results	31
5.1	Evidence Hierarchy and Dataset Identity	31
5.1.1	Canonical Schema and Comparability	32
5.2	Primary Real-Data Experiments	32
5.2.1	Comparable Daily-Demand Benchmark	32
5.2.2	Fine-Grained SKU-Day Demand	34
5.2.3	Real Future-Repeat Prediction	34

5.2.4	Bangladesh Market-Price Benchmark	35
5.3	Controlled Synthetic Comparison	35
5.3.1	Retuned Leakage-Safe Segment Classification	35
5.3.2	Synthetic Churn and Return Risk	36
5.3.3	Synthetic Monthly Forecast Selection	37
5.3.4	Synthetic RFM Clustering	38
5.4	Reproducibility and Validity	38
6	Conclusions	39
6.1	Research Summary	39
6.2	Current Completion Assessment	40
6.3	Future Work Plan	40
6.3.1	Priority 1: Ethics and Primary Data	40
6.3.2	Priority 2: Final Leakage-Safe Experiments	41
6.3.3	Priority 3: Ablation, Scalability and Strategy Value	41
6.3.4	Priority 4: Owner-Centered Evaluation and Hardening	41
6.4	Project Timeline: February 2026–February 2027	41
6.5	Expected Final Outputs	43
	Bibliography	44

List of Figures

3.1	Conceptual operational data model used by the prototype.	18
3.2	Proposed B-SMART end-to-end architecture and feedback loop.	21
3.3	Sequential flow from a business event to outcome tracking.	22
4.1	Implemented local application architecture.	27
5.1	Observed and selected-model predictions on the real Bangladesh demand holdout.	33
5.2	Fixed-holdout WAPE for the shared daily-quantity task; lower is better. .	34
5.3	Development CV and untouched-holdout comparison for tuned segment models.	36
5.4	Precision–recall and ROC curves for the selected synthetic churn model. .	37
5.5	Rolling-origin development results used for synthetic forecast selection. . .	37

List of Tables

2.1	Literature matrix for the current core set of twenty-two papers.	12
2.2	Existing-system limitations and the proposed response.	13
3.1	Evidence tiers and their permitted claims.	17
3.2	Worked reorder example.	23
4.1	Current functional status.	25
4.2	Implementation map.	26
5.1	Current data register and defensible use.	31
5.2	Cross-dataset task comparability matrix.	32
5.3	Comparable fixed-holdout daily-demand results; generated data are a control only.	33
5.4	UCI real future-repeat results.	34
5.5	WAPE on real Bangladesh commodity-price holdouts.	35
5.6	Retuned segment-classification comparison on generated data.	35
5.7	Untouched-holdout risk ranking on generated data.	36
6.1	Progress assessment at 6 September 2026.	40
6.2	Thesis work plan from February 2026 to February 2027.	42

Chapter 1

Introduction

1.1 Motivations

Business intelligence and analytics convert operational records into descriptive, predictive and prescriptive support for decision makers [5]. The opportunity is particularly relevant to small businesses because a modest shop already generates invoices, product movements, mobile-wallet payments, customer histories and supplier transactions. The data often remain separated across paper ledgers, spreadsheets and point solutions, so the owner can answer what happened but not reliably why it happened or what action should follow.

Consider an inventory-based retailer before Eid. The owner must decide how much stock to purchase, which customers are becoming inactive, which product or branch is under-performing and whether a promotion is profitable. Under-forecasting causes stock-outs and lost sales; over-forecasting locks scarce capital in slow stock. A generic monthly chart cannot resolve this decision because it does not combine demand history, current stock, supplier lead time, promotion, calendar context, uncertainty and cash constraints.

Bangladesh adds important design requirements. Research on rural Bangladeshi SMEs found that organizational, technological and environmental factors shape ICT adoption [12], while later work identifies cost, management support, relative advantage, competitive pressure and government support as significant adoption determinants [25]. Studies of mobile payment and MFS use also show that ease, expected benefit and local usage practices matter [10, 22]. Therefore, a usable system should treat local payment channels and business calendars as core data, use low-friction workflows, and explain recommendations in language an owner can act upon.

The approved thesis title is broad, but the implementation scope is deliberately narrowed to Bangladeshi retail and inventory-based SMEs. This makes data requirements, workflows, evaluation targets and business actions concrete while allowing future sector-specific templates.

1.2 Problem Statement

Existing small-business tools commonly stop at transaction recording and static dashboards. Standalone research models, on the other hand, may forecast or classify a public dataset without entering the operational loop where a business records sales, controls stock and acts on a prediction. Three gaps follow:

1. **Fragmented operations and analytics.** Sales, inventory, customer, purchase and payment data are not maintained in one auditable, tenant-isolated system.
2. **Weak scientific validity.** Synthetic or foreign data are sometimes treated as local field evidence; repeated customers cross random train/test splits; post-outcome variables leak into features; and complex models are not compared against simple baselines.
3. **No decision bridge.** A model score is presented without a feasible action, uncertainty, business constraint, local explanation or mechanism for recording outcomes.

The central problem is consequently to design and evaluate one reproducible framework that connects trustworthy data capture, leakage-free analytics and actionable strategy for Bangladeshi SMEs. The work must remain useful on small datasets, must disclose when a baseline is stronger than an AI model, and must not claim production readiness before real local validation.

1.3 Objectives

1.3.1 General Objective

The general objective is to develop and evaluate an AI-powered business analytics system that helps Bangladeshi retail SMEs monitor performance, extract explainable insights and prioritize feasible business actions.

1.3.2 Specific Objectives

- O1. Design a canonical, auditable and tenant-isolated data model covering sales, stock, purchases, returns, customers, suppliers, payments and expenses.
- O2. Build reproducible preprocessing and temporal feature pipelines for demand, future inactivity, segmentation and performance analysis.
- O3. Compare statistical, machine-learning and simple baseline methods using task-appropriate metrics and leakage-safe splits.

- O4. provide global and instance-level explanations while clearly indicating whether a recommendation comes from a trained model or a baseline rule.
- O5. Convert forecasts, risks and KPIs into constraint-aware reorder, retention and operational action rankings.
- O6. Validate the method first on transparent synthetic/public datasets and finally on consenting Bangladeshi SME records and owner-centered evaluation.

1.4 Contributions

The contribution is an integrated framework rather than a claim of a new base classifier. At the present stage, the project contributes:

- **An operational-to-analytical architecture.** The same local system records business events, exports a canonical research dataset, trains models and serves recommendations.
- **Scientific guardrails.** Customer-level or temporal splits, training-only transformations, explicit baseline comparisons, provenance and separation of synthetic, public-real and primary-real evidence are part of the design.
- **A documented leakage case study.** A legacy 94.69% segment score used target-defining CLV. A new development-only tuning protocol and untouched 1,000-customer test produce 69.90% selected-model accuracy; a post-purchase satisfaction feature similarly inflated return prediction.
- **Bangladesh-aware product design.** The schema supports BDT, bKash, Nagad, Bangla QR, local seasonal events and a Bangla-first operational interface.
- **A constraint-aware recommendation layer.** Demand, stock, lead time, consent and cost are evaluated before an action is ranked.
- **Reproducible real-data evidence.** The repository contains a common daily-demand benchmark on UCI and Bangladesh observations, a UCI future-repeat study, WFP/DAM price benchmarks, saved models, audit JSON, figures and tests.

The intended final contributions additionally include a consenting multi-SME dataset protocol, leave-one-business-out evaluation, calibrated uncertainty, owner usability results and a simulated or pilot business-impact comparison.

1.5 Challenges

The principal challenge is data rather than model availability. Public Bangladeshi sources currently provide market prices or a single demand series, but not the joint customer–SKU–stock–transaction history required for the full research questions. Partner recruitment, consent, de-identification and schema reconciliation are therefore on the critical path.

Technical challenges include intermittent SKU demand, few observations for new businesses, changing festivals and prices, missing identifiers, class imbalance, concept drift and the gap between an accurate prediction and a profitable decision. Scientific challenges include leakage, fair baseline selection, uncertainty calibration and transportability from one business to another. Human challenges include digital skill variation, time pressure at the point of sale, trust in model output and explaining uncertainty without technical jargon.

1.6 Scope and Current Report Boundary

This document is a progress report current to 6 September 2026. It covers the project period February 2026 - February 2027. The implemented system, verified tests and executed synthetic/public-data experiments are reported in the past tense. Planned primary SME experiments, user studies and pilot impact evaluation are explicitly reported as future work. This distinction prevents a functioning prototype from being mistaken for a validated commercial product.

1.7 Organization

Chapter 2 defines the terminology, reviews more than twenty relevant works and identifies limitations of existing systems. Chapter 3 formalizes the research questions, data model, algorithms, architecture and evaluation protocol. Chapter 4 documents the implemented platform. Chapter 5 reports current experiments and threats to validity. Chapter 6 summarizes progress and gives the February 2026–February 2027 work plan.

Chapter 2

Related Works

2.1 Background Study

2.1.1 Business Intelligence and Decision Support

Business intelligence (BI) is the set of data, infrastructure, analytical methods and organizational practices used to improve decisions. Its evolution from structured enterprise reporting toward web, mobile and sensor analytics is described by Chen, Chiang and Storey [5]. In this thesis, *descriptive* analytics summarizes revenue, margin and stock; *predictive* analytics estimates future demand or customer inactivity; and *prescriptive* analytics ranks feasible actions.

Operational and analytical functions must be connected but separately governed. An operational transaction requires atomic consistency and an audit trail. A model requires a time-frozen feature view, a reproducible split and versioned artifacts. B-SMART connects the two through a canonical export instead of training directly on mutable application tables.

2.1.2 Bangladeshi SME and Payment Context

This study uses the term SME for the retail-oriented subset of cottage, micro, small and medium enterprises (CMSMEs). Exact regulatory thresholds may change; the operational research population is therefore defined functionally: an inventory-holding Bangladeshi retailer with branch, SKU, sales, purchase and payment records, limited analytics staff and owner-managed decisions.

Bangladesh-specific evidence indicates that the perceived advantage, cost, organizational support, infrastructure and skills surrounding a technology affect adoption [12, 25]. Mobile payments are not an optional integration in this context. SME adoption research identifies performance expectancy, effort expectancy and cost as significant [22], while qualitative

MFS research emphasizes accessibility, mobility and self-service [10]. The data model consequently treats cash, bKash, Nagad, Bangla QR, card and due settlement as first-class choices.

2.1.3 Customer Analytics: RFM, CLV and Segmentation

RFM represents a customer by recency R (how recently the customer purchased), frequency F (how often) and monetary value M (how much). These behavioral features are inexpensive to derive from invoices and often more actionable than demographic attributes. CLV extends the view toward expected future customer value; Fader, Hardie and Lee show that equal RFM scores need not imply equal future value and formalize the relationship using iso-value curves [9].

Supervised segmentation predicts a pre-existing business label. Unsupervised segmentation asks whether coherent groups arise from behavior without such a label. K-Means minimizes within-cluster squared distance,

$$\underset{\{C_k, \mu_k\}_{k=1}^K}{\text{minimize}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \quad (2.1)$$

Because a clustering algorithm always returns clusters, internal validation is essential. The silhouette coefficient compares cohesion with the closest other cluster [23]. A low value should be interpreted as overlap or continuum, not concealed behind attractive segment names.

2.1.4 Demand Forecasting

A demand forecast estimates units sold at a future horizon. ARIMA models linear dependence after differencing; seasonal-naïve forecasting repeats the corresponding prior-season value; tree ensembles learn nonlinear relations among lags, calendar, price and promotion variables. LSTM networks use gates to retain or discard state over long sequences [11], but the extra capacity is not automatically beneficial when a shop has only a short history.

Forecasts must be evaluated chronologically. RMSE emphasizes large errors, MAE preserves the target unit, WAPE aggregates absolute error relative to aggregate demand, and MASE compares error with a naïve scale. Hyndman and Koehler warn that percentage metrics can be undefined or unstable around zero and recommend scaled errors for cross-series comparison [13]. Therefore, final SKU evaluation will report WAPE/MASE with baseline-relative and business-cost measures rather than one convenient metric.

2.1.5 Churn, Future Repeat and Returns

In a non-contractual retail setting, churn is not an observed cancellation. It is defined relative to a cutoff t : a customer has no repeat purchase in the future interval $(t, t + H]$. Features must be frozen at t and the label window must not overlap training and test periods. Returns are similarly predicted only from information known before dispatch; satisfaction recorded after delivery is an outcome, not a valid feature.

Both problems can be imbalanced. A classifier that predicts the majority class may show high accuracy while identifying no useful customer. Precision–recall curves, lift, calibration and expected campaign or return cost are more useful than accuracy alone [24, 26].

2.1.6 Explainability, Leakage and Model Governance

SHAP attributes the difference between a model output and a reference value to input features using a cooperative-game formulation [18]. TreeExplainer provides efficient exact or consistent explanations for common tree ensembles and permits aggregation from local to global behavior [17]. An explanation does not prove causality or fairness; it states how a fitted model used the encoded input.

Data leakage occurs when the training process receives information that would not legitimately be available at prediction time [15]. Leakage may enter through a target-derived field, repeated entity in multiple splits, post-outcome feature or transformation fitted on all data. B-SMART therefore versions features, separates customers or time periods, purges outcome horizons and fits preprocessing on training data only.

2.2 Literature Review

The following review contains twenty-two core papers, exceeding the minimum of twenty requested for the present stage. Each entry states its problem, context, method, main contribution and relevance or limitation.

2.2.1 Bangladesh and Business Analytics Foundations

Chen, Chiang and Storey (2012). The paper organizes BI research into successive generations spanning structured enterprise data, web intelligence and mobile/sensor contexts [5]. It synthesizes more than a decade of publications and identifies research opportunities rather than evaluating one SME implementation. It motivates the breadth of B-SMART, but leaves local adoption, operational integration and field impact to domain-specific studies.

Hoque et al. (2016). This Bangladesh case study investigates the use, impact and adoption of ICT in rural SMEs [12]. It identifies factors that influence adoption in a context underrepresented by mainstream enterprise-systems research. Its contribution is contextual rather than predictive modelling; it supports a low-friction local product design but does not provide retail transaction data.

Shahadat et al. (2023). Using a structured survey of 535 SME managers and PLS-SEM, the authors evaluate technology, organization and environment factors affecting digital adoption [25]. Relative advantage, cost, management support, competitive pressure and government support emerge as important. Survey intent does not demonstrate forecasting accuracy, so B-SMART complements it with transaction-level evaluation.

Rahman and Islam (2023). The authors study mobile-payment adoption among 200 Bangladeshi SMEs using an extended UTAUT model and PLS-SEM [22]. Performance and effort expectancy, social influence, cost and voluntariness significantly influence behavioral intention. The study justifies native local-payment workflows, but it does not address analytics, inventory or longitudinal business outcomes.

Hazra and Priyo (2021). This qualitative study uses affordance theory and interviews to explain how Bangladeshi users achieve access, mobility, self-control and disintermediation through mobile financial services [10]. It contributes rich usage context that purely technical studies omit. Its user-level qualitative setting does not quantify SME model performance, but it informs the system’s payment and usability assumptions.

2.2.2 Customer Analytics and Segmentation

Ngai, Xiu and Chau (2009). The authors screen 900 papers and classify 87 CRM data-mining studies by customer lifecycle dimension and analytical function [21]. Classification and association are prominent, and retention receives substantial attention. The review provides a taxonomy, but predates current explainability and temporal leakage practice; B-SMART joins several functions in one governed pipeline.

Fader, Hardie and Lee (2005). The study links RFM behavior to CLV through Pareto/NBD and gamma-gamma components and validates the approach on CDNOW customers [9]. Iso-value curves demonstrate how different transaction histories can imply comparable future value. The method is principled but assumes a non-contractual purchasing process and does not solve operational inventory decisions.

Kim et al. (2006). Using six months of Korean wireless-service data covering 16,384 customers, the paper develops customer-value dimensions and segment-specific strategies [16]. It shows how current value, potential value and loyalty can support differentiated action. The sector and contractual relationship differ from Bangladeshi retail, limiting direct transfer of labels and features.

Cheng and Chen (2009). This work combines quantitative RFM attributes, K-Means and rough-set theory to classify customer value [7]. It explores three, five and seven value classes and extracts interpretable decision rules. The contribution is a segmentation procedure, while business constraints, temporal validation and post-deployment monitoring remain outside its scope.

2.2.3 Predictive Models and Forecasting

Breiman (2001). Random Forests combine decorrelated tree predictors constructed from bootstrap samples and random feature subsets [1]. The method offers classification, regression, internal error estimation and variable importance, with robustness to noise. It is a strong tabular baseline, though raw importance does not explain an individual business recommendation.

Chen and Guestrin (2016). XGBoost introduces a scalable regularized tree-boosting system with sparsity-aware learning and efficient approximate split finding [6]. The paper reports strong performance across large benchmark tasks. Its engineering strength does not remove the need for leakage-safe data or calibration; in this thesis it is compared with simpler models rather than assumed to win.

Hochreiter and Schmidhuber (1997). The LSTM architecture addresses decaying error flow in recurrent learning through gated memory cells [11]. Experiments on artificial long-lag tasks show that it learns dependencies missed by earlier recurrent methods. Those results do not guarantee accuracy on a 48-month retail series, which motivates the thesis's data-volume rule and naive baseline.

Carbonneau, Laframboise and Vahidov (2008). The authors compare recurrent networks, neural networks and support-vector machines with naive, moving-average and regression forecasts on simulated supply chains and actual Canadian foundry orders [3]. Advanced models perform well, but their advantage over regression is not statistically significant. This directly supports disciplined baseline comparison in small or noisy supply-chain data.

Hyndman and Koehler (2006). This paper evaluates common univariate forecast-accuracy measures and identifies degenerate behavior in several percentage formulations [13]. It proposes MASE as a scale-free alternative suitable across series. The work concerns measurement rather than retail decisions; the present thesis adds WAPE, prediction intervals and inventory cost.

Makridakis, Spiliotis and Assimakopoulos (2018). The M4 competition evaluates point and interval forecasts across 100,000 series [19]. Combinations and hybrid methods dominate, while pure ML entries often fail to beat statistical benchmarks. The heterogeneous series are not Bangladesh-specific, but the result is strong evidence against equating model complexity with forecasting quality.

Makridakis, Spiliotis and Assimakopoulos (2022). The M5 competition focuses on 42,840 hierarchical Walmart retail series and includes explanatory variables and uncertainty forecasts [20]. It establishes a realistic large-scale retail benchmark and highlights hierarchy, intermittency and uncertainty. Walmart’s volume and infrastructure differ sharply from small Bangladeshi shops, leaving a need for low-data, local validation.

2.2.4 Churn, Evaluation Integrity and Explainability

Burez and Van den Poel (2009). The study treats customer churn as an imbalanced classification problem and compares under-sampling, gradient boosting and weighted Random Forests using AUC and lift [2]. It demonstrates that imbalance treatment and metric choice affect conclusions. Its service-industry setting motivates, but does not define, non-contractual retail inactivity.

Verbeke et al. (2012). The authors argue that statistical discrimination alone can select a financially inferior churn model and propose a maximum-profit measure [26]. Telecommunications datasets show gains from choosing both the model and contact fraction by expected profit. B-SMART adopts the decision focus, while requiring SME-specific action cost and consent estimates.

Saito and Rehmsmeier (2015). Through theoretical and empirical comparisons, this paper shows why precision–recall plots reveal performance differences hidden by ROC plots on imbalanced data [24]. Its examples are not retail-specific, but the metric lesson is general. Accordingly, churn and return experiments report PR-AUC, base rate and lift rather than accuracy alone.

Kaufman et al. (2012). The paper formalizes leakage as target information that is illegitimately available to a data-mining model and discusses detection and avoidance [15]. Competition and industrial examples show that leakage can produce convincing but invalid scores. It directly explains the thesis’s CLV, customer-split and post-satisfaction ablations.

Lundberg and Lee (2017). The authors unify several additive feature-attribution methods and introduce SHAP values with consistency properties [18]. The framework supports local explanations for complex models and connects them to global patterns. SHAP is descriptive of a model rather than causal, so B-SMART presents it as supporting evidence rather than a reason to automate an action.

Lundberg et al. (2020). This work develops efficient explanation methods for tree models and shows how many local explanations reveal global model structure [17]. It makes XGBoost and Random Forest behavior more inspectable at serving time. The method still inherits biases and leakage from the fitted model, which is why feature/time audits precede explanation in the proposed framework.

2.3 Literature Matrix

Table 2.1 condenses the core comparison. “Gap” states the limitation most relevant to this thesis, not a claim that the cited work is defective for its own objective.

TABLE 2.1: Literature matrix for the current core set of twenty-two papers.

Ref.	Context/data	Method	Contribution	Gap addressed by B-SMART
[5]	Cross-domain BI literature	Framework and bibliometrics	BI evolution and research agenda	Local SME implementation and measured action loop
[12]	Rural Bangladesh SMEs	Adoption case study	ICT adoption factors	Transaction analytics and impact evaluation
[25]	535 SME managers	PLS-SEM / TOE	Adoption determinants	Longitudinal operational evidence
[22]	200 Bangladesh SMEs	Extended UTAUT, PLS-SEM	Mobile-payment adoption factors	Native payment records linked to analytics
[10]	Bangladesh MFS users	Interviews, affordance theory	Contextual MFS uses	SME workflow and model evaluation
[21]	87 CRM studies	Review taxonomy	CRM/data-mining map	Modern XAI, leakage and integrated system
[9]	CDNOW customer data	Pareto/NBD, gamma-gamma	Formal RFM-CLV link	Operational action constraints
[16]	16,384 Korean telecom users	CLV segmentation	Segment strategies	Non-contractual Bangladesh retail validity
[7]	Customer-value data	RFM, K-Means, rough sets	Interpretable value classes	Temporal evaluation and action tracking
[1]	General benchmarks	Random Forest	Robust ensemble method	Calibration and decision explanation
[6]	Large benchmark tasks	Regularized boosting	Scalable sparse tree learning	Low-data and leakage-safe validation
[11]	Artificial sequences	Gated recurrent network	Long-lag learning	Short retail histories and simple baselines
[3]	Simulated supply chain + foundry orders	ML vs traditional forecasts	Comparative demand evidence	Bangladesh data and inventory utility
[13]	Forecast benchmarks	Metric analysis	MASE and metric cautions	Business cost and interval coverage
[19]	100,000 mixed series	Forecast competition	Hybrid/combination evidence	Small-shop context and actions
[20]	42,840 Walmart series	Hierarchical retail competition	Point and uncertainty benchmark	Low-resource local transportability
[2]	Service churn	Sampling and ensembles	AUC/lift under imbalance	Non-contractual retail label and consent
[26]	Telecom churn datasets	Profit-based evaluation	Links selection to campaign profit	SME costs and measured outcomes
[24]	Imbalanced binary tasks	ROC/PR analysis	PR plots expose minority performance	Lift, calibration and decision cost
[15]	Competitions and practice	Leakage taxonomy	Detection/avoidance framework	Entity/time purging and provenance
[18]	Complex prediction models	Shapley attribution	Unified local explanations	Business-language action explanation
[17]	Tree ensembles	TreeExplainer	Efficient local-to-global XAI	Governance, causal caution and monitoring

2.4 Problem of Existing Systems

The relevant comparison is between capability patterns rather than brand names, because commercial product features and licensing change. Table 2.2 states the recurring limitation and the designed response.

TABLE 2.2: Existing-system limitations and the proposed response.

System/study pattern	Limitation	B-SMART response
Generic POS or book-keeping	Records transactions and totals but lacks temporal prediction and action evaluation.	Canonical event ledger feeds forecasts, risks and ranked actions.
Generic ERP for larger firms	Configuration, cost and vocabulary may exceed a small owner-managed shop's capacity.	Narrow retail MVP, Bangla-first daily workflows and local payment choices.
Static BI dashboard	Explains past KPIs but provides no forward uncertainty or feasible next step.	Forecast, risk, constraint check, explanation and action rank are connected.
Standalone forecasting paper	Evaluates a series without stock, supplier, budget or deployment workflow.	Forecast enters reorder logic using stock, lead time, safety stock and cost.
Synthetic retail study	Demonstrates a pipeline but cannot establish field validity.	Synthetic evidence is labelled; public-real and future primary-real results are separate.
Randomly split customer model	The same customer or a future-derived attribute can appear on both sides of evaluation.	Grouped or chronological splits, purged horizons and training-only transforms.
Accuracy-only churn model	Majority prediction can appear successful and still contact no useful customer.	PR-AUC, ROC-AUC, Brier score, lift and campaign utility.
Black-box recommendation	Gives a score without evidence, uncertainty or owner-readable reason.	SHAP/supporting factors, confidence source and explicit constraints.
Foreign single global model	Misses Bangladesh calendar/-payment context and masks business heterogeneity.	Local context features plus per-business and leave-one-business-out tests.
Unmonitored deployed model	Does not record version, drift, owner response or later outcome.	Versioned artifacts, provenance, audit log and planned outcome tracking.

2.5 Research Gap and Positioning

Prior research establishes the value of BI, RFM/CLV, tree ensembles, temporal forecasting, churn ranking and explainability. Bangladesh-focused adoption work explains why cost,

usability, local payments and organizational readiness matter. However, no reviewed work jointly provides a Bangladesh-oriented SME operations system, leakage-safe temporal models, constraint-aware action ranking, local explanations, transparent evidence tiers and measured owner outcomes.

B-SMART is positioned in this intersection. Its novelty is the governed sequence from business event to evidence-backed action, not a new classifier. The current prototype validates feasibility and exposes methodological failure modes. The final thesis must still demonstrate transportability and usefulness with consenting Bangladeshi SMEs before making claims about real business impact.

Chapter 3

Proposed Methodologies

3.1 Research Design

The thesis follows a design-science and empirical evaluation approach. A software artifact is designed to solve a practical decision problem, while its analytical components are tested under reproducible protocols. The research has four evidence stages: (i) synthetic data for safe development and failure-mode analysis, (ii) public real data for external method validation, (iii) primary consenting Bangladeshi SME data for the main claims, and (iv) owner-centered usability and decision-impact evaluation.

The unit of forecasting is branch-SKU-day or branch-SKU-week, depending on data density. The unit of customer-risk modelling is a customer snapshot at a historical cutoff. The unit of strategy evaluation is an eligible action whose cost, risk and operational feasibility are known.

3.2 Research Questions and Hypotheses

RQ1.

Do Bangladesh-aware calendar, price, promotion and market features improve branch-SKU demand forecasts over seasonal and statistical baselines?

RQ2.

How accurately and reliably can historical customer snapshots predict no repeat purchase in the following 60 or 90 days?

RQ3.

Does a forecast-driven reorder policy reduce stock-outs and holding cost compared with the participating owner's existing policy?

RQ4.

Do concise Bangla action explanations improve task completion, confidence and perceived usability for SME owners?

The associated hypotheses are:

H1.

A model with local calendar, price, promotion and market features produces lower out-of-time WAPE/MASE than its otherwise identical ablation.

H2.

A temporally evaluated customer-risk model achieves PR-AUC above the outcome base rate and lift at 10% above 1.0 on unseen future snapshots.

H3.

Under identical demand sequences and constraints, forecast-driven replenishment reduces total stock-out plus holding cost relative to the current-policy baseline.

H4.

In a counterbalanced owner study, the explanation condition improves task completion or confidence without a harmful increase in completion time.

These hypotheses will be evaluated per business and in aggregate. Failure to reject a null hypothesis will be reported as a result, not repaired by changing the metric after inspection.

3.3 Target Population and Data Acquisition

The target population is inventory-based retail SMEs in Bangladesh. The minimum primary target is three to five businesses from a comparable vertical, with 12–24 months of invoice lines, at least 50,000 lines in total, approximately 500 SKUs and 2,000 pseudonymous customers. A smaller first partner with six months of records may be used to validate collection and import before the full study.

TABLE 3.1: Evidence tiers and their permitted claims.

Tier	Source	Permitted use	Prohibited inference
1	Consenting SME invoices and stock records	Bangladesh Primary forecasting, churn, strategy and usability claims	None beyond sampled firms without external validation
2a	WFP/DAM market prices	Bangladesh Real local context and price forecasting benchmark	SKU sales, customer behavior or SME revenue
2b	UCI Online Retail II, UK [4]	Real transaction-pipeline and future-repeat validation	Bangladesh purchasing behavior
2c	Bangladesh retailer daily demand [14]	Real local quantity-forecast validation	Revenue, customer or multi-SKU claims
3	32,000-row generated Bangladesh-context dataset	Development, leakage case study, demonstration	Field effectiveness or real customer behavior

Each import stores a provenance manifest: organization, date range, export system, collector, received date, checksum, row count, permission/license, schema version, transformations, exclusions and known limitations.

3.4 Ethics, Privacy and Data Governance

University ethics approval and a signed data-sharing agreement precede primary collection. Names, phone numbers, e-mail addresses, national IDs, account numbers and wallet numbers are excluded from research tables. Each organization receives a different salted pseudonymous customer identifier; any re-identification mapping remains under the business’s control outside the repository.

The agreement records purpose, permitted researchers, retention deadline, publication form and deletion procedure. Only aggregated or separately reviewed de-identified outputs may be published. Customer contact recommendations require recorded consent. Tenant identifiers are enforced in application queries and foreign keys so one organization’s data cannot enter another’s model or dashboard.

3.5 Canonical Data Model

The operational model contains Organization, Branch, Product, Customer, Supplier, SalesOrder, SalesOrderItem, Payment, InventoryBalance, StockMovement, PurchaseOrder,

Expense, LedgerEntry, SalesReturn, Refund, AuditLog and ImportBatch entities. The research export converts these events to a canonical invoice-line table, while the synthetic research pipeline also demonstrates a star schema with customer, product, date, business, marketing and logistics dimensions around a transaction fact.

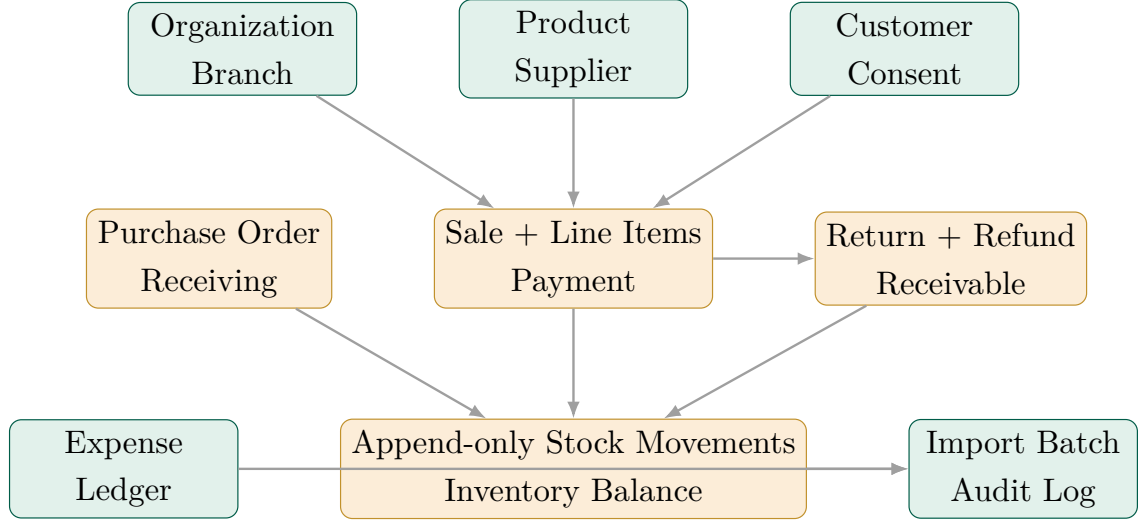


FIGURE 3.1: Conceptual operational data model used by the prototype.

3.6 Formal Problem Formulation

For business b , branch l , product p , customer c and time t , define

$$D = \{\text{Sales, Inventory, Purchase, Expense, Customer, Payment, Context}\}. \quad (3.1)$$

3.6.1 Demand Prediction

$$\hat{y}_{b,l,p,t+h} = f(X_{\text{history}}, X_{\text{price}}, X_{\text{promotion}}, X_{\text{calendar}}, X_{\text{market}}), \quad (3.2)$$

where the output is expected units at horizon h . Quantile models or calibrated residual intervals will provide a plausible demand range, not only a point.

3.6.2 Future Customer Inactivity

$$q_{c,t} = P(\text{no purchase in } (t, t+H] \mid X_{c,t}), \quad (3.3)$$

where every feature in $X_{c,t}$ is computed using timestamps no later than t .

3.6.3 Strategy Optimization

For an eligible action $a \in A$,

$$U(a) = E[\Delta \text{Profit}(a)] - \text{Cost}(a) - \lambda \text{Risk}(a), \quad (3.4)$$

subject to budget, storage, current/incoming stock, supplier lead time, minimum order quantity and customer consent. The system returns the top feasible actions with model/baseline provenance and supporting factors.

3.7 Preprocessing and Historical Feature Construction

The pipeline validates keys, timestamps, monetary identities and stock movement reconciliation before modelling. Duplicates and cancellations remain auditable; they do not silently enter revenue. Missingness is profiled. Imputation, encoding and scaling are fitted only to training periods or training customers.

Demand features include lagged units, rolling mean/variance, day-of-week, month, festival indicators, price, discount, stock-out flag, promotion and external market price where available. Customer snapshots include recency, frequency, monetary value, order-value statistics, product/category breadth, return rate, discount behavior, tenure and purchase rate. The outcome window is separated from the observation window by a purge interval for future-repeat modelling.

3.8 Analytical Models

3.8.1 Segmentation and Classification

K-Means on scaled RFM features discovers candidate behavioral groups. Elbow and silhouette diagnostics inform K , but business interpretability is reported separately from natural separation. Logistic Regression, Random Forest [1] and XGBoost [6] predict existing labels only after excluding label-derived CLV and grouping records by customer.

3.8.2 Forecasting

The baseline ladder begins with last-value and seasonal-naive forecasts, followed by statistical ARIMA/SARIMA, gradient boosting on lagged features, and LSTM only when sequence volume supports it. This ordering implements evidence from forecast competitions that complex models do not consistently dominate naive/statistical benchmarks [19, 20].

3.8.3 Churn and Return Risk

Logistic Regression, Random Forest and XGBoost are compared. Churn is a future-repeat outcome; return risk uses pre-dispatch features only. Class weights and threshold selection may be tuned on validation data. Final selection uses PR-AUC, calibration, lift and estimated action value rather than accuracy.

3.8.4 Explainability

SHAP values provide global rankings and local contributions for tree models [17, 18]. Explanations are transformed into short supporting statements. They identify associations used by the model and are never worded as causal conclusions.

3.9 The B-SMART Sequential Algorithm

Algorithm 1: B-SMART Strategy Recommendation

Input: Business data D , horizon H , constraints C , model versions M , actions A

Output: Ranked actionable recommendations R

- 1 Validate schema, timestamps, tenant ownership and missingness;
 - 2 Build a time-aware feature snapshot X_t ;
 - 3 Predict SKU demand and uncertainty intervals;
 - 4 Predict future customer inactivity probabilities;
 - 5 Calculate KPIs and abnormal deviations;
 - 6 Generate candidate actions from forecasts, risks and KPIs;
 - 7 **foreach** $a \in A$ **do**
 - 8 Estimate benefit, cost and risk;
 - 9 Check a against constraints C ;
 - 10 $U(a) \leftarrow E[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\text{Risk}(a)$;
 - 11 Remove infeasible actions and rank the remainder by $U(a)$;
 - 12 Attach explanation, confidence source and supporting evidence;
 - 13 Return the top- K actions and record later owner response/outcome;
-

For N transaction lines and P candidate actions, validation and feature aggregation are $O(Nd)$ for d inspected fields, and action filtering/ranking is $O(P \log P)$. Training cost depends on the selected model: for example, a forest with T trees has an approximate $O(TN \log N)$ construction cost under balanced trees. Serving reuses versioned artifacts, so online work is feature retrieval, model inference and $O(P \log P)$ ranking rather than retraining.

3.10 Architecture and Sequential Flow

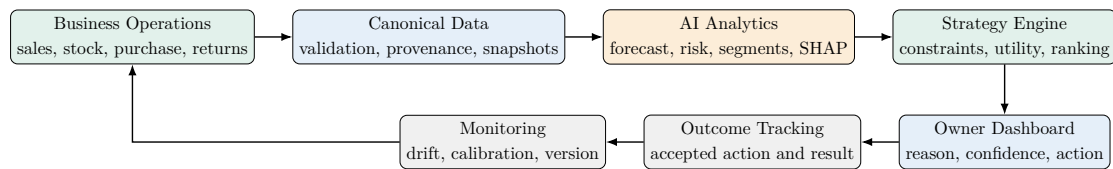


FIGURE 3.2: Proposed B-SMART end-to-end architecture and feedback loop.

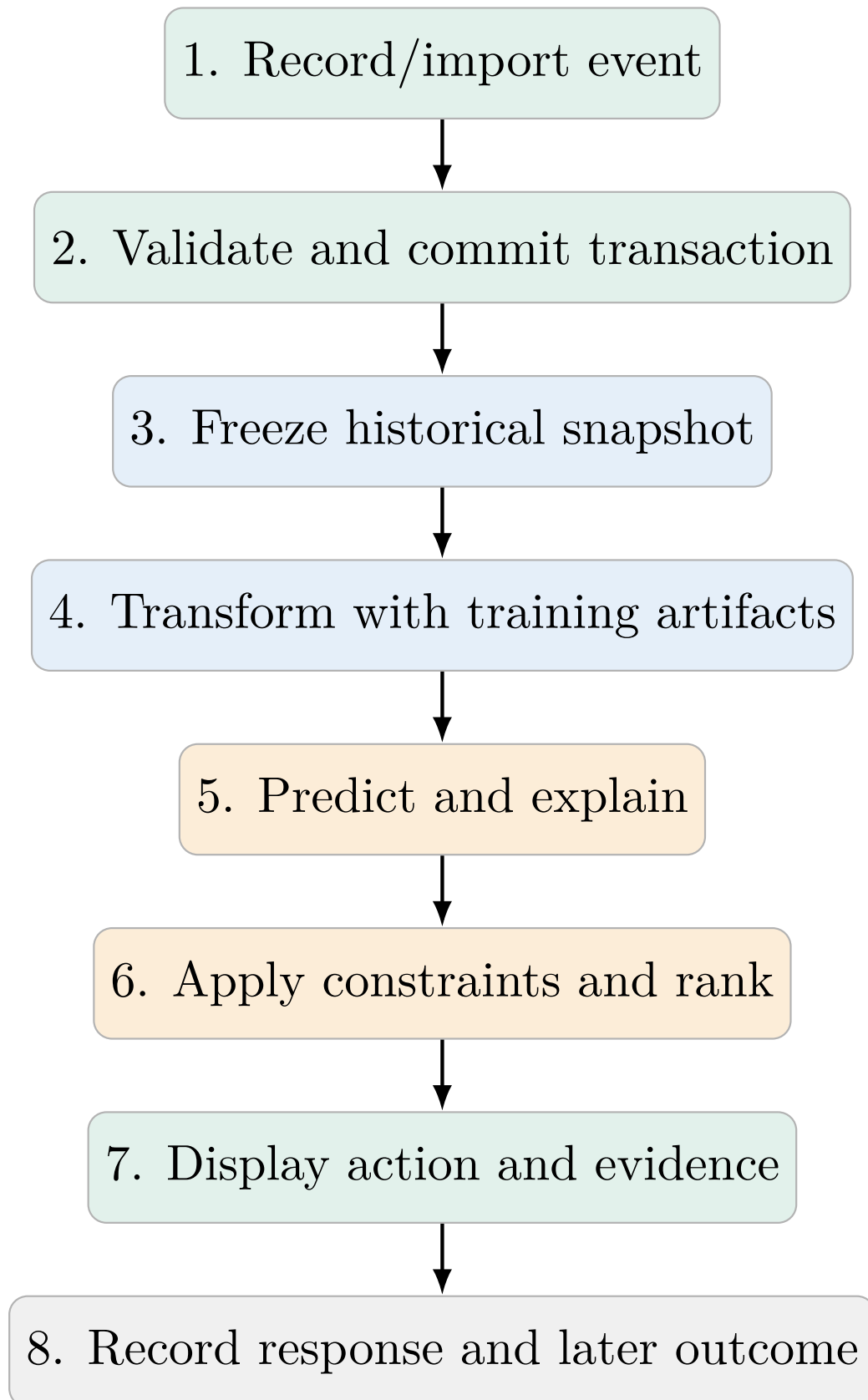


FIGURE 3.3: Sequential flow from a business event to outcome tracking.

3.11 Worked Recommendation Example

Suppose a shop has the state in Table 3.2. Values are a privacy-safe illustration, not an experimental result.

TABLE 3.2: Worked reorder example.

Product	Last 7-day sales	Stock	Lead time	Forecast
Soybean Oil	30	12	4 days	34
Rice 5 kg	18	40	3 days	20
Sugar 1 kg	25	8	5 days	29

For Soybean Oil with safety stock six units and no incoming order,

$$\text{Reorder} = \max(0, 34 + 6 - 12) = 28 \text{ units.} \quad (3.5)$$

The interface should state: “Expected seven-day demand is 34 units (with an uncertainty range shown by the model). Current stock is 12 and supplier lead time is four days. Order approximately 28 units today to cover forecast demand and safety stock.” If no trained real-data model exists, the same card is labelled as a baseline recommendation.

3.12 Evaluation Protocol

3.12.1 Forecasting Metrics and Splits

Models are trained on earlier observations and tested on the final untouched period. Rolling-origin evaluation will be used when data volume allows. For actual values y_i and forecasts $\hat{y}_i$,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.6)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3.7)$$

$$\text{WAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}. \quad (3.8)$$

MASE, interval coverage, stock-out units, holding units and total simulated cost will supplement these measures. Calendar, price/promotion, market context and business-specific modelling will be removed one group at a time for ablation.

3.12.2 Classification and Ranking Metrics

Customer-risk models report PR-AUC, ROC-AUC, Brier score, calibration plots, lift at 10% and expected campaign profit. Segment classifiers report accuracy, macro-F1, confidence intervals and paired McNemar tests. All entity-level tasks keep an entity on one side of a split.

3.12.3 Clustering Evaluation

Candidate K values are compared using inertia, silhouette and stability under resampling. Cluster profiles are then checked for size, RFM differences and actionability. Commercially named clusters are not treated as natural kinds when silhouette or stability is weak.

3.12.4 Scalability, Usability and Business Impact

The scalability plan covers 100, 500, 1,000, 10,000, 100,000 and 1,000,000 synthetic transaction rows, measuring generation, import, preprocessing, training, prediction latency, API throughput, peak memory, artifact size and database query time. Market-basket minimum-support tests are deferred until an association-rule component is implemented; no placeholder values will be used.

Owner usability will measure task completion, completion time, error count, confidence and a standard usability questionnaire, with and without explanation cards. Reorder impact will compare identical demand sequences under the existing policy and B-SMART policy, reporting stock-outs, holding cost and total utility.

Chapter 4

Implementation

4.1 Implementation Status

As of 6 September 2026, the repository contains a complete local thesis MVP. It is not represented as a finished commercial ERP. The operational, analytical and strategy layers execute locally, and current safeguards distinguish a trained real-data model from a transparent baseline. External sign-in, public deployment, fiscal integration, native mobile/offline synchronization and automated campaigns are intentionally deferred.

TABLE 4.1: Current functional status.

Component	Status	Implemented boundary
Business operations	Complete MVP	Catalogue, branch, customer, supplier, stock, sales, purchase, return, expense and ledgers
Data intake	Complete MVP	Durable CSV/XLSX import, validation, checksum/provenance and canonical sales export
Research analytics	Complete MVP	Segmentation, forecasting, churn, return, leakage ablation and SHAP artifacts
Real-data training	Complete MVP	Chronological demand pipeline and purged future-repeat pipeline
Strategy layer	Complete MVP	Constraint-aware reorder and consent-safe retention ranking; baseline/model provenance
API and dashboard	Complete MVP	FastAPI routes plus React research and Bangla-first operational pages
Automated verification	Passing	Six backend tests pass; production frontend builds
Primary SME validation	Pending	Consenting Bangladeshi invoice/stock data and field evaluation required
Association-rule module	Not implemented	Minimum-support scalability experiment remains out of current results
Public production deployment	Deferred	OAuth, managed database, monitoring, billing and hardware integration outside current MVP

4.2 Technology Stack

The backend uses Python 3.12, FastAPI, Pydantic, SQLAlchemy and Alembic. SQLite supports the local prototype, with foreign-key enforcement; the configuration layer permits a production database later. The analytical stack comprises pandas, NumPy, SciPy, scikit-learn, XGBoost, statsmodels, TensorFlow, SHAP, Matplotlib and Seaborn. Models and preprocessing objects are persisted with pickle, Joblib or the native Keras format as appropriate.

The frontend uses React, React Router, Recharts and Vite. A HashRouter permits the static production build to be served without server-side route rewriting. Research views remain technical and English-oriented, while daily operational views are Bangla-first. The current local database is `bsmart.db`; schema migrations are versioned under `migrations/`.

4.3 Repository and Layered Architecture

The implementation separates domain transactions, analytical service code and presentation. Table 4.2 maps major code to its responsibility.

TABLE 4.2: Implementation map.

Module or directory	Responsibility
<code>api/domain_models.py</code>	Multi-tenant operational schema and audit/provenance entities
<code>api/commerce_routes.py</code>	Products, inventory adjustments and atomic sales
<code>api/finance_routes.py</code>	Purchases, receiving, payments, returns, refunds, expenses and ledgers
<code>api/directory_routes.py</code>	Branches, staff, customers and suppliers
<code>api/data_import_routes.py</code>	Durable imports, validation, canonical export and model-training trigger
<code>api/analytics_routes.py</code>	Tenant-scoped KPI and constraint-aware recommendations
<code>predict.py</code>	Loads fixed research artifacts and provides inference/service functions
<code>upload_service.py</code>	Anonymous dataset mapping, analysis and dataset-specific future-repeat training
<code>ml/real_pipeline.py</code>	Chronological real-data demand and future-repeat training
<code>ml/serving.py</code>	Loads organization-specific demand artifacts for strategy recommendations
<code>ml/market_prices.py</code>	Bangladesh commodity-price benchmarks and market-context table
<code>ml/real_data_validation.py</code>	UCI and Bangladeshi public-real method validation
<code>frontend/src/pages/</code>	Operational and research-mode application screens
<code>tests/</code>	Authentication, tenant/database, commerce and analytics invariants

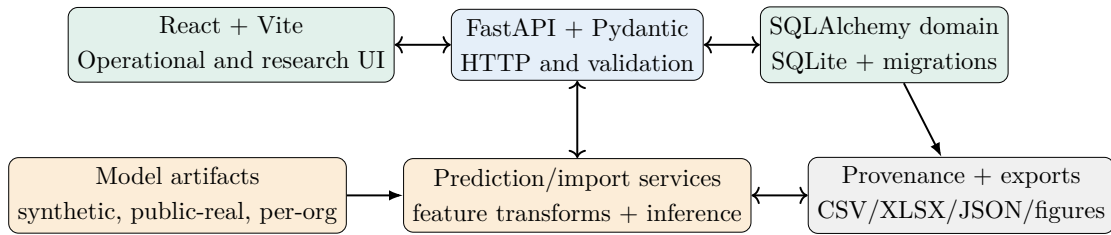


FIGURE 4.1: Implemented local application architecture.

4.4 Operational Layer

4.4.1 Tenant and Branch Isolation

An organization contains branches and memberships. Domain rows carry an organization identifier, and request dependencies resolve the active membership before a query or mutation. Duplicate SKUs are rejected within one tenant while separate tenants may use the same SKU text. The current authentication layer supports local prototype identity and optional JWT verification; Google or other external OAuth is not implemented.

4.4.2 Catalogue, Inventory and Sales

A product stores SKU, category, unit, cost, selling price, barcode and reorder level. Inventory adjustment creates an append-only stock movement and updates the materialized balance. Sale creation validates stock and records the order, line items, payments and stock decrements within one database transaction. Supported settlement methods include cash, bKash, Nagad, Bangla QR, card and due sales.

4.4.3 Purchasing, Returns and Accounts

Purchase orders support partial or full receiving; received lines update stock and supplier payable state. A sale return can optionally restock goods and can create a refund or receivable adjustment. Expenses and payable/receivable ledger entries feed a 30-day operational dashboard. These flows create the data needed for future demand and business-impact evaluation instead of requiring an unrelated research spreadsheet.

4.5 Durable Data Intake

The anonymous research upload flow parses CSV/XLSX columns, suggests mappings, cleans transactions and provides overview, forecast, segmentation, product and future-repeat analysis. The operational import flow is durable: it writes the source file, checksum,

row count and validation status to an ImportBatch record. The canonical export provides the exact schema expected by the real-data training pipeline.

Import validation checks tenant, branch, invoice, line, SKU and timestamp keys; monetary identities; return/cancellation treatment; missingness; and cross-tenant collisions. Phone data are hashed for the operational customer directory and are not included in research features.

4.6 Analytical Pipeline

The original numbered research scripts form a reproducible sequence:

0. `00_normalize_dataset.py`: convert the flat synthetic CSV to six dimensions and one transaction fact, then create a verified merge.
1. `01_preprocessing.py` and `01b_customer_features.py`: clean, encode, engineer RFM/calendar features and create leakage-safe customer splits.
2. `02_classification.py` and `02b_classification_v2.py`: retain the legacy result for ablation and train the honest customer-level XGBoost, Random Forest and Logistic Regression comparison.
3. `03_clustering.py`: evaluate candidate K values and fit K-Means on RFM profiles.
4. `04_forecasting.py`: compare LSTM, ARIMA(2,1,2) and a seasonal-naive forecast on the same holdout.
5. `05_shap_analysis.py`: generate global/local SHAP artifacts; current legacy plots are interpreted as a leakage audit where relevant.
6. `06_final_report.py`: aggregate metrics and figures.
7. `07_return_prediction.py` and `08_churn_prediction.py`: evaluate pre-dispatch return and future inactivity risk using imbalance-aware metrics.

The separate real-data path uses chronological splits and organization-specific artifacts. A trained demand model is stored at

```
artifacts/real/<organization-id>/demand_model.joblib.
```

Serving first looks for this artifact; when it does not exist or lacks history for a SKU, a clearly labelled 28-day demand-rate baseline is used.

4.7 Strategy Optimization Layer

The recommendation endpoint reads only the active tenant's canonical records. For each product it combines observed or modelled demand rate, current stock, reorder level and lead time to generate a candidate reorder. For customers it uses recency/value information and contact consent to generate retention candidates. The action includes estimated utility, reason, method and a **confidence** field with value **model** or **baseline**.

This provenance is a scientific and user-interface control. The UI does not rename a heuristic as AI while data are insufficient. Automated campaigning is not performed; the system ranks consented customers for human review.

4.8 API and User Interface

The API exposes health, research dashboard, customer, prediction, anonymous upload, operational commerce, directory, finance, data import, training and strategy routes. The React application contains the following page groups:

- **Daily operations:** setup, sales, products, inventory, purchases, returns, directory, accounts and the 30-day operational dashboard.
- **Strategy:** constraint-aware action cards with method/confidence.
- **Research mode:** overview, dataset upload, what-if analysis, segments, customers, forecast, model report and real-data validation.

The current design keeps evaluator-facing model detail in English and owner-facing daily tasks in Bangla. A final usability study will test whether this split and the explanation cards improve decisions.

4.9 Verification and Build Status

On 6 September 2026, `python -m pytest -q` completes successfully with six passing tests. The tests cover token behavior, wrong-audience rejection, tenant-scoped SKU/s-stock invariants, duplicate-SKU rejection, atomic sale/stock decrement and operational analytics. The set is a high-value smoke suite rather than complete coverage; failure-path, concurrency, migration and browser tests remain to be expanded.

The command `npm run build` transforms 611 modules and completes the production build successfully. The main minified JavaScript asset is about 769.61 kB (219.15 kB gzip), triggering a chunk-size warning. Route-level dynamic imports and code splitting

are therefore an implementation task before public deployment, but the warning does not prevent the local thesis demonstration.

4.10 Reproducibility

The Python 3.12 environment is used because TensorFlow support is not available for the system's newer default interpreter. Random seeds are set where supported, but minor LSTM variation remains possible because early stopping and backend operations are not fully deterministic. Deterministic baseline results and stored artifacts preserve the ranking. The repository build guide lists the commands and output map.

Chapter 5

Experimental Results

5.1 Evidence Hierarchy and Dataset Identity

The experimental evidence is real-data first. Three public real sources are used for the claims they can support, while the 32,000-line Bangladesh-context dataset generated by `01_generate_dataset.py` is retained only as a controlled software and modelling baseline. It is not presented as collected business evidence. UCI Online Retail II contains genuine UK invoices [4]; the Mendeley series contains genuine daily demand from a Bangladeshi retailer [14]; and the WFP/DAM source contains observed Bangladeshi market prices rather than store sales [27]. The World Bank FAT survey provides national technology adoption context but is not used to train the prediction models [8].

TABLE 5.1: Current data register and defensible use.

Dataset	Usable observations	Status	Permitted task
UCI Online Retail II	1,044,415 demand lines; 805,620 identified lines	Real, UK	Daily demand and 90-day future-repeat validation
Bangladesh retailer demand	1,826 days, one product	Real, Bangladesh	Quantity forecasting only
WFP/DAM food prices	24,500 filtered benchmark rows	Real, Bangladesh	Monthly market-price forecasting/context
World Bank FAT 2019	Survey meta-data/context	Real, Bangladesh	Motivation and technology-adoption context only
Generated B-SMART data	32,000 lines; 4,996 customers	Synthetic	Controlled segment, churn, return and monthly forecast experiments
Partner SME invoices	Not yet collected	Pending	Final Bangladesh customer and multi-SKU field validation

5.1.1 Canonical Schema and Comparability

The sources are not artificially forced to contain the same information. The adapters map observed fields to a canonical schema with sale date, item, quantity, unit value and pseudonymous customer identifier. A placeholder is used only when a downstream validator requires a field and the placeholder is excluded from analysis. For example, the Bangladesh demand adapter sets one constant anonymous customer and unit price 1.0; consequently, neither customer nor revenue results are computed from it.

TABLE 5.2: Cross-dataset task comparability matrix.

Source	Daily quantity	Customer repeat	Price series	Synthetic labels
UCI Online Retail II	Yes	Yes	Observed unit price	No
Bangladesh retailer series	Yes	No customer ID	No	No
WFP/DAM	No sales quantity	No	Yes	No
Generated B-SMART	Yes	Designed target	Designed values	Yes

Raw accuracy is therefore not compared across incompatible targets. Daily demand is compared with weighted absolute percentage error (WAPE), MAE, RMSE and R^2 ; future-repeat uses PR-AUC, ROC-AUC, Brier score and top-decile lift; multiclass segment prediction uses accuracy and macro-F1. This follows the forecast-metric cautions of Hyndman and Koehler [13] and the recommendation to emphasize precision–recall behaviour for imbalanced classification [24].

5.2 Primary Real-Data Experiments

5.2.1 Comparable Daily-Demand Benchmark

For an apples-to-apples experiment, the two real sources and generated control are reduced to observed total quantity per calendar day. Features use only prior values: lags 1, 2, 3, 7, 14 and 28; shifted rolling means and standard deviations; and calendar encodings. The final 15% of dates form a fixed chronological holdout. Extra Trees, Random Forest, histogram gradient boosting, XGBoost, squared- and absolute-error variants, and three naive baselines are ranked by mean WAPE over four expanding-window folds inside the first 85%. The chosen family is then fitted once and evaluated on the holdout. This is a

rolling one-day-ahead forecast, meaning the previous day’s observed quantity is available when predicting the next day.

TABLE 5.3: Comparable fixed-holdout daily-demand results; generated data are a control only.

Dataset	Selected model	Days	WAPE	MAE	R^2	Holdout
UCI Online Retail II	XGBoost L1	107	30.37%	6,676.38	0.437	25 Aug–9 Dec 2011
Bangladesh retailer demand	Extra Trees	270	17.49%	4.18	0.411	6 Apr–31 Dec 2017
Generated B-SMART control	Extra Trees L1	215	25.28%	13.86	-0.042	31 May–31 Dec 2024

The complement $100 - \text{WAPE}$ is 69.63% for UCI, 82.51% for the Bangladeshi series and 74.72% for the generated control, but the report does not call this “accuracy” because WAPE is an error ratio, not a classification success rate. The local series is the stronger result; it is still limited to one retailer and one product. UCI is more volatile and includes closure days and bulk orders, so its larger error is not hidden. The synthetic control has negative R^2 , showing that even a moderate WAPE does not mean its temporal variation is explained well.

The objective-alignment audit added absolute-error learners after the initial squared-error run. UCI WAPE remained near 30% (29.04% before the audit and 30.37% for the deterministic CV-selected L1 candidate). This stability argues against a simple model-code defect or an honest quick accuracy gain. Since candidate expansion followed inspection of the earlier holdout, these numbers are development evidence rather than a final confirmatory test; a future period or partner-SME dataset must provide that test.

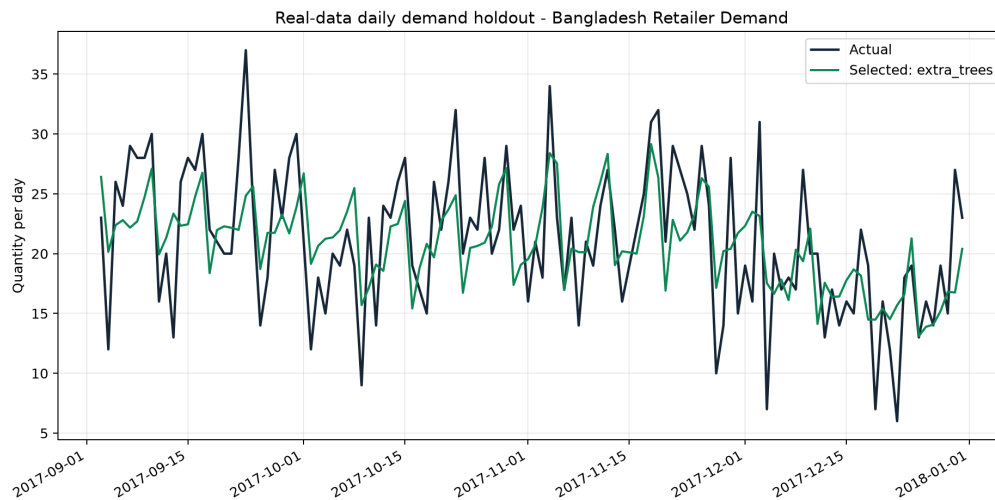


FIGURE 5.1: Observed and selected-model predictions on the real Bangladesh demand holdout.

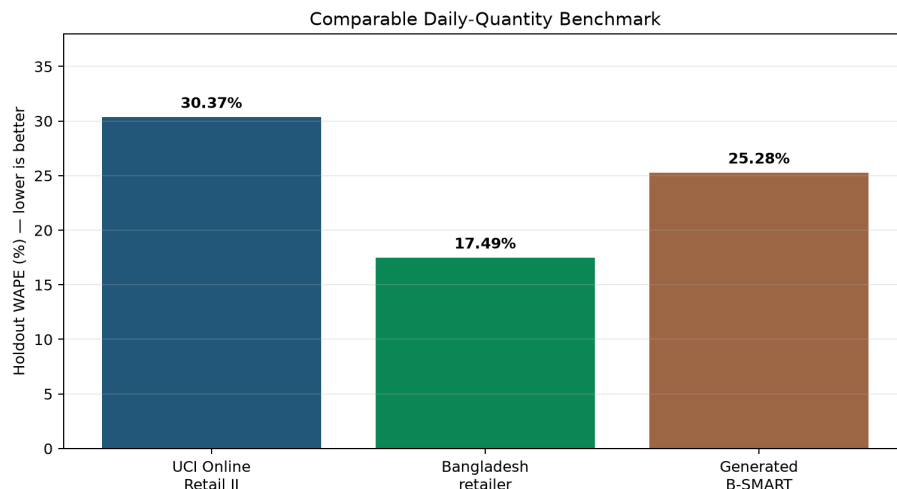


FIGURE 5.2: Fixed-holdout WAPE for the shared daily-quantity task; lower is better.

5.2.2 Fine-Grained SKU-Day Demand

A harder operational test preserves separate item-day rows. UCI is restricted to the top 40 SKUs by total quantity to avoid creating a multi-million-row full daily grid. Histogram gradient boosting obtains WAPE 1.138 versus 1.225 for the seven-day seasonal-naive baseline, a 7.1% relative error reduction; however, WAPE above 1 means total absolute error exceeds total observed demand. This is not deployment-ready performance. On the single-product Bangladesh series, the same pipeline obtains WAPE 0.190 versus 0.238, a 20.1% relative reduction. The aggregate and SKU-day views answer different questions and are both kept.

5.2.3 Real Future-Repeat Prediction

Only UCI contains customer identifiers and enough history for this task. Snapshots use 180 days of past activity to predict whether the customer returns within the following 90 days. Training and test cutoffs are chronological, and a 90-day purge prevents training outcomes from overlapping test features. All 5,881 customers are eligible; the test contains 11,014 snapshots with positive rate 0.5356.

TABLE 5.4: UCI real future-repeat results.

Model	ROC-AUC	PR-AUC	Brier	Lift@10%
Random Forest	0.7487	0.7954	0.2034	1.77x
Logistic Regression	0.7460	0.7911	0.2070	1.77x

Random Forest wins by the predeclared Brier-score criterion. PR-AUC 0.795 is well above the 0.536 prevalence baseline, and the top-ranked decile contains about 1.77 times

the positive rate of random contact. This validates the method on genuine transactions but not transfer to Bangladeshi customers.

5.2.4 Bangladesh Market-Price Benchmark

Four WFP/DAM commodity series have enough history for a final 12-month test window. WAPE is shown as a percentage below. These are national market prices, not SME sales or inventory records.

TABLE 5.5: WAPE on real Bangladesh commodity-price holdouts.

Commodity	Months	Seasonal	Last value	SARIMA	Winner
Rice (coarse)	216	14.23%	3.14%	16.16%	Last value
Wheat flour	212	8.66%	8.55%	7.85%	SARIMA
Lentils (masur)	199	14.24%	6.97%	21.99%	Last value
Oil (palm)	197	6.41%	5.58%	23.31%	Last value

Simple last-value forecasting wins three series and SARIMA wins one. Gradient boosting, also evaluated, is worse on three of four. These results reinforce the forecasting literature’s warning that complex learning algorithms do not automatically defeat carefully chosen baselines [19, 20].

5.3 Controlled Synthetic Comparison

5.3.1 Retuned Leakage-Safe Segment Classification

The synthetic comparison uses a new seed (20260905) and a previously untouched 1,000-customer holdout. All preprocessing, univariate feature selection, hyperparameter sampling and family choice occur within the other 3,996 customers using five-fold stratified cross-validation. Customer ID and CLV are excluded because CLV defined the segment label. Four model families were tuned; XGBoost was selected before opening the holdout because it had the highest development CV accuracy (70.19%).

TABLE 5.6: Retuned segment-classification comparison on generated data.

Model family	Development CV accuracy	Holdout accuracy	Macro-F1
XGBoost (selected)	70.19%	69.90%	71.52%
Extra Trees	69.62%	70.10%	71.99%
Random Forest	69.37%	68.60%	70.59%
Histogram Gradient Boosting	69.17%	69.30%	71.28%

The selected XGBoost holdout accuracy is 69.90% (bootstrap 95% CI [67.0%, 72.6%]), balanced accuracy 70.33%, and majority-class accuracy 43.6%. Extra Trees happens to score 0.2 percentage point higher on this holdout, but switching after seeing the test would be test-set selection; the reported primary result therefore remains 69.90%. This corrects the legacy 94.69% score that leaked target-defining CLV and repeated customers, consistent with leakage-avoidance guidance [15].

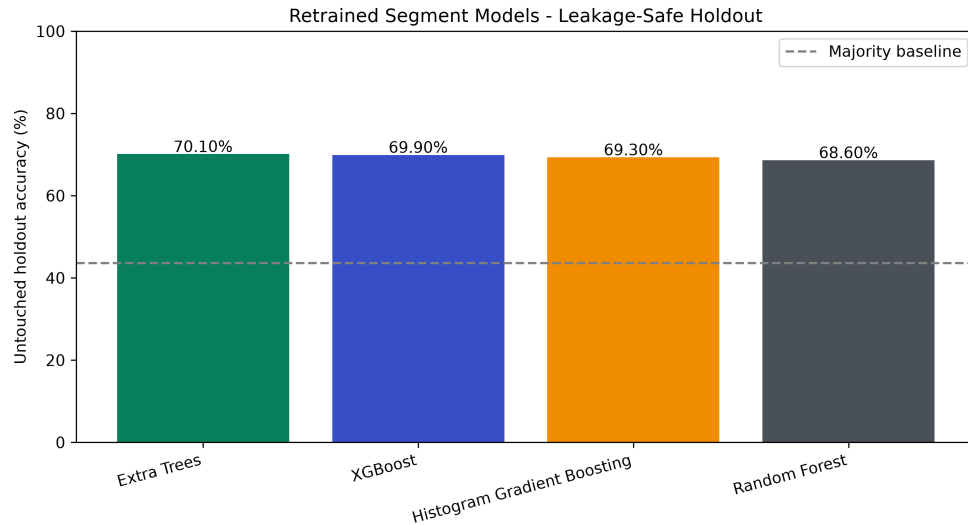


FIGURE 5.3: Development CV and untouched-holdout comparison for tuned segment models.

5.3.2 Synthetic Churn and Return Risk

Binary models are selected by development cross-validated PR-AUC and their decision thresholds by out-of-fold F1. The churn experiment excludes recency, RFM and other label-derived fields. Return prediction is customer-group disjoint and excludes delivery status, reason, satisfaction and all other post-outcome variables.

TABLE 5.7: Untouched-holdout risk ranking on generated data.

Task	Selected	Base	ROC	PR	Lift@10%	F1
Churn	Logistic Regression	0.202	0.722	0.421	2.67x	0.427
Return	Logistic Regression	0.068	0.556	0.083	1.40x	0.145

Churn ranking is useful within the simulator; return ranking remains weak and should not drive automated action. Majority prediction would produce over 93% return “accuracy” while detecting no returned purchases, illustrating why accuracy alone is misleading under imbalance [2, 24].

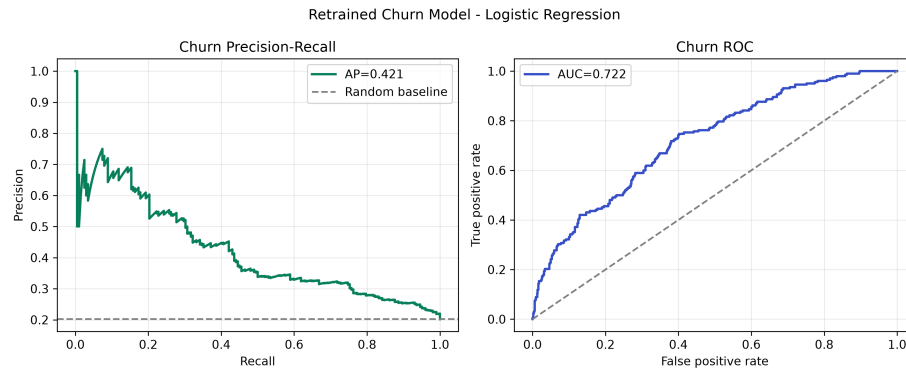


FIGURE 5.4: Precision–recall and ROC curves for the selected synthetic churn model.

5.3.3 Synthetic Monthly Forecast Selection

The last six of 48 months are untouched. Fourteen candidate specifications are ranked by mean WAPE across three expanding-window, six-month validation folds inside the first 42 months. Additive Holt–Winters is selected at 7.43% mean CV WAPE, but obtains 13.46% WAPE (12.33% MAPE) on the final holdout. The predefined seasonal-naive diagnostic obtains 8.13% WAPE (6.60% MAPE) on that holdout. Because the baseline was not selected by the same CV rule, it is reported as a diagnostic winner rather than substituted after test inspection.

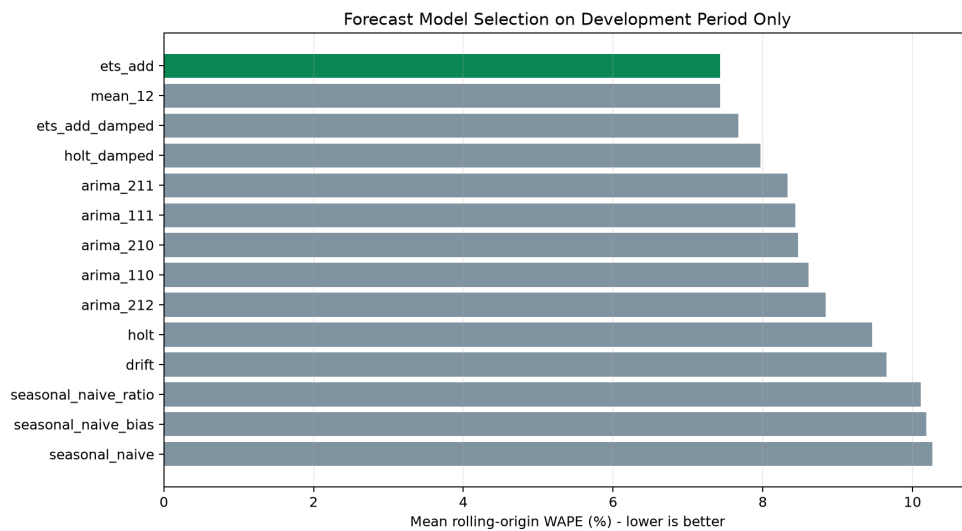


FIGURE 5.5: Rolling-origin development results used for synthetic forecast selection.

The failure of the selected model to transfer from three small validation windows demonstrates uncertainty from a 48-point series. It argues for more history and simple baselines rather than tuning until the final six months look favourable.

5.3.4 Synthetic RFM Clustering

Candidate silhouette scores are 0.2837 for $K = 2$ and 0.1725 for $K = 4$. Therefore, $K = 2$ is the statistical choice by silhouette. The four-profile view (2,691 low-engagement, 1,131 VIP-platinum, 877 moderate-spender and 297 high-value customers) is retained only as a business-facing summary requested by the application design; it is not claimed to be the naturally optimal partition. Silhouette interpretation follows Rousseeuw [23].

5.4 Reproducibility and Validity

Every table above is backed by a saved JSON or pickle artifact and a timestamped run log. Raw public files have SHA-256 hashes in the project documentation. The bibliography audit resolves 27 references against Crossref, DataCite or an official publisher/repository page. The audit is machine-readable at `output/reference_verification.json`.

Construct validity remains limited because inactivity is a future-repeat proxy, not contractual churn, and WAPE measures forecast error rather than profit. External validity is limited by one Bangladeshi product series and one UK transaction source. Internal validity is improved by chronological/group splits, purging, training-only selection and label-derived feature exclusions, but only a prospective partner-SME study can validate business impact in Bangladesh. The present results support a functioning research prototype and a disciplined evaluation protocol; they do not support claims of nationwide deployment readiness.

Chapter 6

Conclusions

6.1 Research Summary

This thesis addresses the gap between recording small-business transactions and making defensible operational decisions from them. The proposed B-SMART framework connects an auditable retail operations layer, leakage-safe predictive analytics, explanation, business constraints and action ranking. The system is Bangladesh-oriented through its payment methods, calendar/context features, currency, Bangla-first daily interface and primary data plan.

The current implementation is a working local MVP. It covers branch and tenant isolation, products, inventory, sales, purchases, returns, finance, customers, suppliers, durable imports, canonical exports, model training, an API, a web dashboard and strategy recommendations. The codebase passes its current six-test backend suite and produces a valid production frontend build.

The strongest current evidence comes from public real data. A task-aligned daily-demand benchmark selects a model within each development period and obtains fixed-holdout WAPE of 17.49% on the Bangladesh retailer series and 30.37% on UCI Online Retail II. UCI future-repeat prediction reaches PR-AUC 0.795 and ROC-AUC 0.749. On WFP/DAM Bangladesh commodity prices, last-value forecasts win three of four series and SARIMA wins one. Complexity must therefore be earned by out-of-time evidence.

The controlled synthetic results provide a second methodological lesson. A legacy segment score of 94.69% is invalid because CLV nearly reconstructs the label. With a new untouched 1,000-customer test, training-only feature selection and model-family choice, selected XGBoost obtains 69.90% accuracy and 71.52% macro-F1. Synthetic churn reaches 2.67x top-decile lift, while return prediction remains weak. Negative results are retained rather than concealed.

The report does not conclude that the system improves real SME performance yet. That claim requires consenting invoice and stock histories, calibrated per-business tests, an owner study and strategy impact measurement. The present contribution is a technically complete platform and a scientifically honest path to that evidence.

6.2 Current Completion Assessment

TABLE 6.1: Progress assessment at 6 September 2026.

Work package	Assessment	Evidence / exit condition
Problem and scope	Complete for current stage	Approved title, retail SME scope and four research questions
Core literature review	Complete for current stage	Twenty-two synthesized papers, 27 verified references and literature matrix; expand for final
Operations MVP	Complete for local thesis use	End-to-end commerce, inventory, finance, import and audit flows
Analytics MVP	Complete for current stage	Segmentation, forecast, churn, return, SHAP and leakage experiments
Strategy MVP	Complete for current stage	Reorder/retention ranking with baseline/model provenance
Public-real validation	Complete for current stage	Comparable UCI/Bangladesh demand, UCI future-repeat and WFP/DAM benchmark artifacts
Primary SME acquisition	Critical pending item	Ethics approval, agreements and 3–5 comparable businesses
Final empirical study	Pending	Temporal/leave-one-business-out tests, ablations, calibration and costs
Owner usability and impact	Pending	Counterbalanced tasks plus policy simulation or pilot
Final thesis and defence	In progress	Verified report, final figures, similarity audit and presentation

6.3 Future Work Plan

6.3.1 Priority 1: Ethics and Primary Data

The immediate task is to finalize the target retail vertical, ethics submission, consent form, data-sharing agreement and data dictionary. Three to five partners will be recruited. Each file will pass provenance, privacy, key, monetary, stock/return reconciliation, missingness and date-range checks before modelling.

6.3.2 Priority 2: Final Leakage-Safe Experiments

Demand models will use rolling or final-period evaluation. The baseline ladder will include last-value, seasonal-naive and statistical methods before gradient boosting or sequence models. Future-repeat snapshots will be horizon-purged and evaluated with PR-AUC, ROC-AUC, Brier score, calibration and lift. Results will be reported per business and under leave-one-business-out transfer. Corrected SHAP artifacts will be generated only from the final leakage-safe models.

6.3.3 Priority 3: Ablation, Scalability and Strategy Value

Ablations will remove calendar, market-price, price/promotion and customer features and will compare global with business-specific models. Scalability runs will extend through one million synthetic rows with repetitions and uncertainty. The reorder policy will be evaluated on identical demand paths against owner or standard baselines using stock-out and holding costs. Association-rule minimum-support experiments will be conducted only if market-basket analysis is implemented within the final scope.

6.3.4 Priority 4: Owner-Centered Evaluation and Hardening

Representative users will complete operational and decision tasks with and without concise explanation cards. The study will record completion, time, errors, confidence and usability. Engineering work includes broader failure and concurrency tests, frontend code splitting, backup/restore, calibration/drift views and versioned outcome tracking. OAuth, statutory accounting and native offline mobile support remain optional extensions unless required for the study.

6.4 Project Timeline: February 2026–February 2027

Table 6.2 uses C for completed work, A for active work at the status date and P for planned work. A blank cell means the work package is not scheduled for that month. Because the status date is early September, September contains both completed/current and active work packages.

TABLE 6.2: Thesis work plan from February 2026 to February 2027.

Work package	Feb 26	Mar 26	Apr 26	May 26	Jun 26	Jul 26	Aug 26	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Feb 27
Problem, proposal and scope	C	C											
Background and core literature	C	C	C	C				A	P	P	P	P	
Synthetic data and initial pipeline		C	C	C	C	C							
Leakage audit and corrected models				C	C	C	C						
Operations/API/frontend MVP			C	C	C	C	C	C					
Public-real data validation							C	C					
Ethics, partner agreements and pilot import								A	P				
Primary SME data collection/cleaning									P	P	P		
Final temporal models and ablations										P	P	P	
Scalability and strategy simulation											P	P	
Owner usability/pilot evaluation											P	P	
Thesis writing, audit and revision								A	P	P	P	P	P
Defence preparation and submission												P	P

C completedA active at status dateP planned

6.5 Expected Final Outputs

By February 2027 the project aims to deliver:

1. a reproducible thesis and source repository with a 40+ work literature review and verified references;
2. a documented de-identified primary SME dataset or, if recruitment is constrained, a transparent account of collected scope and limitations;
3. leakage-safe demand and future-repeat benchmarks with calibration, ablation and per-business transfer analysis;
4. a tested constraint-aware strategy policy with stock-out/holding-cost simulation or pilot outcomes;
5. usability results for owner-facing explanations;
6. a hardened local demonstration, final diagrams, experiment artifacts and defence presentation; and
7. a publication-ready manuscript focused on the strongest verified contribution, most likely leakage-safe low-data decision support for Bangladeshi retail SMEs.

The final submission will retain negative results and evidence labels. If primary data volume is insufficient for a deep model or a field impact claim, the thesis will narrow the claim instead of expanding the conclusion beyond the evidence.

Bibliography

- [1] BREIMAN, L. Random forests. *Machine Learning* 45 (2001), 5–32.
- [2] BUREZ, J., AND VAN DEN POEL, D. Handling class imbalance in customer churn prediction. *Expert Systems with Applications* 36, 3 (2009), 4626–4636.
- [3] CARBONNEAU, R., LAFRAMBOISE, K., AND VAHIDOV, R. Application of machine learning techniques for supply chain demand forecasting. *European Journal of Operational Research* 184, 3 (2008), 1140–1154.
- [4] CHEN, D. Online retail ii. UCI Machine Learning Repository, 2012. CC BY 4.0; accessed 6 September 2026.
- [5] CHEN, H., CHIANG, R. H. L., AND STOREY, V. C. Business intelligence and analytics: From big data to big impact. *MIS Quarterly* 36, 4 (2012), 1165–1188.
- [6] CHEN, T., AND GUESTRIN, C. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2016), pp. 785–794.
- [7] CHENG, C.-H., AND CHEN, Y.-S. Classifying the segmentation of customer value via RFM model and RS theory. *Expert Systems with Applications* 36, 3 (2009), 4176–4184.
- [8] CIRERA, X., COMIN, D., AND CRUZ, M. Bangladesh – the firm adoption of technology survey, 2019. World Bank Microdata Library, 2026. Nationally representative firm-level technology adoption survey metadata; accessed 6 September 2026.
- [9] FADER, P. S., HARDIE, B. G. S., AND LEE, K. L. RFM and CLV: Using iso-value curves for customer base analysis. *Journal of Marketing Research* 42, 4 (2005), 415–430.
- [10] HAZRA, U., AND PRIYO, A. K. K. Mobile financial services in bangladesh: Understanding the affordances. *The Electronic Journal of Information Systems in Developing Countries* 87, 3 (2021), e12166.
- [11] HOCHREITER, S., AND SCHMIDHUBER, J. Long short-term memory. *Neural Computation* 9, 8 (1997), 1735–1780.

- [12] HOQUE, M. R., SAIF, A. N. M., ALBAR, A. M., AND BAO, Y. Adoption of information and communication technology for development: A case study of small and medium enterprises in bangladesh. *Information Development* 32, 4 (2016), 986–1000.
- [13] HYNDMAN, R. J., AND KOEHLER, A. B. Another look at measures of forecast accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688.
- [14] JAHIN, M. A., AND SHAHRIAR, A. Supply chain demand forecasting dataset of bangladeshi retailer. Mendeley Data, Version 1, 2024. CC BY 4.0; accessed 6 September 2026.
- [15] KAUFMAN, S., ROSSET, S., PERLICH, C., AND STITELMAN, O. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data* 6, 4 (2012), 15:1–15:21.
- [16] KIM, S.-Y., JUNG, T.-S., SUH, E.-H., AND HWANG, H.-S. Customer segmentation and strategy development based on customer lifetime value: A case study. *Expert Systems with Applications* 31, 1 (2006), 101–107.
- [17] LUNDBERG, S. M., ERION, G., CHEN, H., DEGRAVE, A., PRUTKIN, J. M., NAIR, B., KATZ, R., HIMMELFARB, J., BANSAL, N., AND LEE, S.-I. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* 2 (2020), 56–67.
- [18] LUNDBERG, S. M., AND LEE, S.-I. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30 (2017), pp. 4765–4774.
- [19] MAKRIDAKIS, S., SPILIOTIS, E., AND ASSIMAKOPOULOS, V. The M4 competition: Results, findings, conclusion and way forward. *International Journal of Forecasting* 34, 4 (2018), 802–808.
- [20] MAKRIDAKIS, S., SPILIOTIS, E., AND ASSIMAKOPOULOS, V. The M5 competition: Background, organization, and implementation. *International Journal of Forecasting* 38, 4 (2022), 1325–1336.
- [21] NGAI, E. W. T., XIU, L., AND CHAU, D. C. K. Application of data mining techniques in customer relationship management: A literature review and classification. *Expert Systems with Applications* 36, 2 (2009), 2592–2602.
- [22] RAHMAN, M. A., AND ISLAM, M. A. Identifying the factors affecting adoption of mobile payment system by small and medium sized enterprises in bangladesh. *Dhaka University Journal of Business Studies* 42, 2 (2023), 47–73. Official journal article page; accessed 6 September 2026.
- [23] ROUSSEEUW, P. J. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20 (1987), 53–65.

- [24] SAITO, T., AND REHMSMEIER, M. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE* 10, 3 (2015), e0118432.
- [25] SHAHADAT, M. M. H., NEKMAHMUD, M., EBRAHIMI, P., AND FEKETE-FARKAS, M. Digital technology adoption in SMEs: What technological, environmental and organizational factors influence SMEs’ ICT adoption in emerging countries? *Global Business Review* (2023), 1–27.
- [26] VERBEKE, W., DEJAEGER, K., MARTENS, D., HUR, J., AND BAESENS, B. New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European Journal of Operational Research* 218, 1 (2012), 211–229.
- [27] WORLD FOOD PROGRAMME, AND DEPARTMENT OF AGRICULTURAL MARKETING, GOVERNMENT OF BANGLADESH. WFP food prices database – bangladesh. Humanitarian Data Exchange, 2026. CC BY-IGO; accessed 6 September 2026.